Proposal: Gym Exercise Recognition

Goal
Identify exercises such as squats, deadlifts, and bench presses, including transitions within a video, with minimal manual annotation.

Training Workflow

1. **Extract features:** Use frozen DINOv2 to generate frame embeddings, preserving their temporal order. Cache embeddings to reduce training cost.
2. **Discover actions:** Cluster short sequences into candidate action groups using feature similarity and temporal consistency.
3. **Review clusters:** Inspect representative clips, assign exercise names, and merge or correct groups. Include rest and unknown activities.
4. **Train a TCN:** Use PyTorch to train a Temporal Convolutional Network on the reviewed pseudo-labels, predicting an exercise at each time step.
5. **Generate segments:** Smooth predictions and output exercise labels with start and end timestamps.

Validation
Evaluate on a small, manually annotated holdout set. Measure per-exercise F1, segment accuracy, and processing cost. Split data by participant or recording to prevent leakage.

Main Risk
Clusters may reflect camera angles or equipment instead of exercises. Validate clustering quality before training the TCN; introduce sparse labels if needed.

Deliverable
A pipeline that converts gym videos into timestamped exercise segments.